POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Data Communication (New)

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time :3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Simplex, Half Duplex and Full Duplex mode of data transmission. Briefly explain digital communication system with the help of general block diagram. 8
   b) Differentiate between different types of transmission modes. Briefly Explain RS- 232C. 7
2. a) Define Signal. Briefly explain periodic and non-periodic signals, energy and power signals, deterministic and random signals, continuous and discrete time signals. 7
   b) Define unit step signal with necessary mathematical expression and graphical illustration. Check the following system for linearity, causality, static & dynamic and time invariance. 8

$$y(n) = Ax(n) + B$$

3. a) Compare and contrast between OSI and TCP/IP with necessary figures. 7
   b) Define guided and unguided transmission media. State the advantages of optical fiber over other guided media. 8
4. a) Why burst error is difficult to handle than single error? Explain why CRC is better than parity method of error detection. 7

OR

Explain various transmission impairments of a communication system. 8

   b) Compare and contrast between lossless and lossy compression. 8
5. a) Explain the importance of flow control. Briefly explain sliding window flow control mechanism. 7
   b) Explain the multiplexing technique applied in digital telephony with their significance, application and multiplexing hierarchy. 8
6. a) Is switching and routing same? Differentiate between message switching, circuit switching and packet switching. 7
   b) Differentiate between PSK, ASK and FSK modulation techniques with necessary diagram. 7

OR

Encode the Bit Stream 011010001 using the following scheme.

i. NRZ – I
ii. NRZ _ L
iii. Manchester
iv. Differential Manchester

2×5

7. Write short notes on: **(Any two)**
   a) Cellular Telephony
   b) Point to Point protocol
   c) QPSK

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